

PROJECT PROPOSAL (MMPP-001)

ON

A STUDY ON CUSTOMER AWARENESS AND

ADOPTION OF UPI-BASED DIGITAL

PAYMENT SYSTEMS AMONG BANK

ACCOUNT HOLDERS IN MAHARASHTRA



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INTRODUCTION

The way people manage and transfer money has changed significantly over the last decade. In India, the shift from cash-based transactions to digital payments has been one of the most remarkable economic transformations in recent times. At the heart of this transformation lies the Unified Payments Interface, commonly known as UPI — a real-time payment system developed by the National Payments Corporation of India (NPCI) and launched in the year 2016.

UPI was built on a simple but powerful idea — to allow any bank account holder in India to send and receive money instantly using just a smartphone, without the need for cash, cheque books, or even knowledge of the recipient's bank account number. A Virtual Payment Address (VPA) or a simple QR code scan is all that is needed to complete a transaction within seconds.

The growth of UPI since its launch has been extraordinary. According to the Reserve Bank of India's Payment System Report, UPI transactions grew from just 92 crore in the financial year 2017-18 to over 17,221 crore in 2024, recording a compounded annual growth rate of 89.3 percent in volume. The total value of UPI transactions also rose from Rs. 5.86 lakh crore in 2018 to Rs. 246.83 lakh crore in 2024. By 2024, UPI alone accounted for 83 percent of all digital payment transactions in India, up from just 34 percent in 2019. India now processes nearly 48.5 percent of all real-time digital payment transactions in the world, making it a global leader in this space.

This rapid growth has been supported by several enabling factors. The Government of India's Digital India initiative, the Jan Dhan Yojana for financial inclusion, the widespread availability of affordable smartphones, and the expansion of internet

connectivity even in semi-urban and rural areas have collectively created a strong foundation for digital payment adoption across different sections of society.

Despite this impressive growth at a national level, the ground reality at the customer level tells a more complex story. While millions of Indians are actively using UPI on a daily basis, there still exists a considerable section of bank account holders who remain either unaware of the full range of UPI features, or who are hesitant to adopt it due to concerns about security, technical difficulties, or simply a lack of proper understanding. This gap between the availability of UPI and its actual informed adoption among bank customers is a matter worth studying closely.

Maharashtra, being one of India's most economically active and financially developed states, presents an interesting study setting. With a large and diverse banking customer base spread across urban, semi-urban, and rural areas, Maharashtra offers a realistic picture of where India stands in terms of UPI awareness and adoption at the ground level.

The present study is an attempt to examine the level of awareness about UPI-based digital payments among bank account holders in Maharashtra, the factors influencing their adoption behaviour, and the challenges they face in using these services. The findings of this study are expected to provide useful insights for banks, financial institutions, and policymakers who are working towards deepening digital financial inclusion in India.

RATIONALE OF THE STUDY

While India has emerged as a global leader in real-time digital transactions, the macro-level statistics often overshadow the ground-level complexities of financial technology adoption. The Unified Payments Interface (UPI) has penetrated urban retail markets rapidly, yet its acceptance and structured usage among the broader base of bank account holders across diverse demographics remain uneven. This study is undertaken based on the following key rationale:

1.Bridging the Awareness-Adoption Gap: Possessing a bank account and a smartphone does not automatically translate to the informed usage of digital payment systems. There is a critical gap between basic awareness (knowing UPI exists) and functional adoption. This research aims to identify why this gap persists.

2.Investigating Socio-Demographic Variance: Adoption patterns are heavily influenced by age, education level, income brackets, and occupational backgrounds. Understanding how these factors influence a customer's willingness to transition from traditional banking to smartphone-based transactions is crucial for developing targeted financial inclusion strategies.

3.Addressing Security and Trust Barriers: Digital financial fraud, phishing scams, and transaction failures act as severe psychological barriers for traditional bank customers. This study seeks to evaluate the specific risk perceptions and technical anxieties that cause resistance or hesitation among account holders in Maharashtra.

4.Regional Context and Policy Relevance: Maharashtra presents a unique socioeconomic mix of highly commercial urban centers and extensive semi-urban/rural banking populations.

LITERATURE REVIEW

To establish a solid theoretical and empirical foundation for this study, existing research on digital banking, technology adoption frameworks, and UPI systems in India has been reviewed. The literature is broadly categorized into three thematic areas:

3.1 Theoretical Frameworks for Technology Adoption

The academic shift toward understanding digital payment systems heavily relies on established behavioral models.

- **Technology Acceptance Model (TAM):** Widely utilized in banking research, TAM posits that **Perceived Usefulness (PU)** and **Perceived Ease of Use (PEOU)** are the primary determinants of a user's intention to adopt a new technology. In the context of UPI, PEOU is exceptionally high due to simplified interfaces like QR code scanning, which directly accelerates behavioral intention.
- **Unified Theory of Acceptance and Use of Technology (UTAUT/UTAUT2):** Researchers have modified traditional models to include factors such as **Social Influence**, **Facilitating Conditions** (availability of high-speed internet and smartphones), and **Price Value** (zero or minimal transaction charges on UPI).
- **Diffusion of Innovation (DOI) Theory:** Recent literature applying DOI theory to Indian fintech indicates that UPI's adoption is heavily driven by its **Relative Advantage** over traditional NEFT/RTGS banking and its **Observability**—seeing peers transact instantly creates a strong network effect that prompts quick trial and permanent adoption among tech-hesitant bank account holders.

3.2 Key Drivers of UPI and Digital Payment Adoption

Extensive empirical studies conducted between 2020 and 2026 highlight several critical factors that encourage consumers to adopt smartphone-based transaction methods:

- **Convenience and Speed:** Studies reveal an overwhelmingly high positive correlation between transaction convenience and user satisfaction. The ability to complete peer-to-peer (P2P) and peer-to-merchant (P2M) transfers instantly without inputting complex IFSC codes or waiting for beneficiary activation acts as the single largest driver across all demographics.
- **Cost Effectiveness:** Unlike traditional credit cards or merchant payment gateways that levy convenience fees, the zero-fee structure mandated for standard UPI transactions makes it highly appealing to lower-income brackets and small-scale retail consumers, promoting mass-level financial inclusion.
- **Demographic Triggers:** Empirical surveys consistently show a sharp age polarization; consumers aged between 18 and 35 demonstrate an immediate affinity for mobile payments, whereas older demographics require external validation or structural support from family members or bank staff to complete their initial onboarding.

3.3 Barriers and Structural Resistance

Despite skyrocketing macro-level transaction volumes, a distinct subset of literature focuses on the systemic issues causing anxiety and complete non-adoption among traditional bank account holders:

- **Security Anxieties and Trust Deficit:** Security concerns remain the most significant roadblock. The proliferation of digital financial fraud, remote-device-

control phishing scams, and clone apps acts as a psychological barrier. Studies confirm that a lower propensity to adopt mobile banking is directly tied to acute privacy concerns and fear of losing hard-earned savings.

- **Technical Infrastructure Vulnerabilities:** Server downtimes, frequent transaction timeouts, and delayed auto-reversals of debited amounts create severe friction. For middle-to-low-income consumers, a single stuck transaction can cause severe cash-flow disruptions, leading to a total loss of trust in the digital pipeline and a subsequent return to cash reliance.
- **The Literacy-Capability Gap:** Researchers distinguish between basic digital literacy and actual financial capability. While a consumer might know how to run entertainment or communication applications on a smartphone, navigating UPI PIN generation, setting transaction limits, or locating official grievance portals within banking applications presents a steep learning curve.

3.4 Identification of the Research Gap

While existing literature extensively documents national UPI transaction volumes and broad consumer sentiments in major metropolitan hubs, there is a distinct lack of granular empirical evidence focusing on the mid-tier consumer segment in socio-demographically diverse states like Maharashtra. Most studies either focus entirely on urban merchant adoption or general financial inclusion in purely rural blocks.

This study aims to bridge this specific gap by investigating the **mid-spectrum bank account holders** in Maharashtra—examining how varying levels of individual awareness, socio-demographic backgrounds, and personal risk perceptions interact to dictate true long-term adoption versus situational or forced usage.

OBJECTIVES OF THE STUDY

The primary goal of this research is to evaluate the depth of digital payment adoption at the grassroots level among bank account holders. To achieve this, the study is structured around the following specific, measurable objectives:

- **To identify** the level of awareness regarding various UPI features, safety protocols, and grievance redressal mechanisms among bank account holders in Maharashtra.
- **To analyze** the impact of socio-demographic factors (such as age, education level, monthly income, and occupation) on the adoption rate of UPI-based digital payment systems.
- **To evaluate** the primary drivers (such as transactional convenience, speed, zero-fee structure, and peer influence) that motivate customers to choose UPI over traditional banking channels.
- **To determine** the critical barriers and risk perceptions (including fear of digital fraud, server downtimes, transaction timeouts, and complex interfaces) that cause resistance or hesitation among non-users and cautious users.
- **To examine** the degree of customer satisfaction regarding the current infrastructure, transactional reliability, and dispute resolution speed provided by banking UPI applications.

HYPOTHESIS OF THE STUDY

To empirically validate the objectives of this research, the following null and alternative hypotheses have been formulated for statistical testing:

Hypothesis 1 (Demographic Factors & Adoption Rate)

- H₀: There is no significant relationship between the age of bank account holders and their adoption rate of UPI-based digital payment systems.
- H₁: There is a significant relationship between the age of bank account holders and their adoption rate of UPI-based digital payment systems.

Hypothesis 2 (Education Level & Awareness)

- H₀: The level of awareness regarding UPI safety protocols and grievance mechanisms is independent of the formal education level of the bank account holders.
- H₁: The level of awareness regarding UPI safety protocols and grievance mechanisms is significantly dependent on the formal education level of the bank account holders.

Hypothesis 3 (Income Brackets & Transaction Barriers)

- H₀: Risk perceptions and technical anxieties (such as fear of fraud and server timeouts) do not vary significantly across different monthly income brackets of consumers.
- H₁: Risk perceptions and technical anxieties (such as fear of fraud and server timeouts) vary significantly across different monthly income brackets of consumers.

Hypothesis 4 (Occupational Background & Usage Frequency)

- H₀: There is no significant difference in the frequency of UPI usage among bank account holders coming from different occupational backgrounds (salaried, business, student, retired).
- H₁: There is a significant difference in the frequency of UPI usage among bank account holders coming from different occupational backgrounds (salaried, business, student, retired).

SCOPE OF THE STUDY

The scope of this study defines the boundaries, parameters, and specific focus areas within which the research will be conducted. To maintain a realistic and scientifically manageable focus, the scope is categorized into the following dimensions:

- **Geographical Scope:** The study is strictly confined to the state of **Maharashtra, India**. Data collection will target bank account holders residing in selected urban, semi-urban, and rural clusters within the state to capture a comprehensive regional perspective on digital payment patterns.
- **Demographic Scope:** The target population for this research consists of individual **bank account holders** who own a smartphone. This includes a diverse demographic mix across different age groups (18 years and above), formal education levels, occupational backgrounds (salaried, self-employed, students, and retired individuals), and various monthly income brackets.
- **Conceptual and Operational Scope:** The study specifically focuses on the **Unified Payments Interface (UPI)** ecosystem. It evaluates customer awareness levels regarding operational features, transaction safety protocols, and grievance redressal mechanisms. It further investigates the behavioral drivers of adoption and the psychological or technical barriers causing resistance. Other digital banking channels like internet banking, corporate RTGS/NEFT, or credit card networks are outside the scope of this study.
- **Temporal Scope:** The study reflects consumer behavioral trends, transaction experiences, and infrastructure reliability data relevant to the current period of **2024 to 2026**.

RESEARCH METHODOLOGY

This section outlines the structured framework utilized to conduct the empirical study. In accordance with rigorous academic research standards, it details the design, setting, sampling framework, data collection tools, and statistical analysis plan.

7.1 Research Design

The study adopts a **Descriptive and Analytical Research Design** utilizing a cross-sectional survey method. This design is selected to systematically capture, document, and analyze the existing patterns of awareness, adoption drivers, and structural barriers among bank account holders without manipulating the study environment.

7.2 Study Setting

The operational setting for this research is the state of **Maharashtra, India**. To ensure the data reflects a true socio-economic mix, the setting is divided into two primary zones:

- **Zone A (Urban/Semi-Urban Hubs):** Selected lead bank branches and public commercial spaces across major municipal corporations (e.g., Pune and Nagpur regions), capturing high-density retail transaction environments.
- **Zone B (Rural/Semi-Urban Clusters):** Selected Agricultural Produce Market Committee (APMC) market yards and regional rural bank (RRB) access points within the peri-urban fringes of these districts, capturing traditional cash-reliant customer bases transitioning to digital platforms.

7.3 Sampling Technique and Sample Size Justification

To address the limitations of arbitrary sample selection, the sampling process is structured via a strict two-tier framework:

- **Sampling Method:** A **Multi-Stage Stratified Random Sampling** technique is applied. In the first stage, the target regions are stratified into distinct urban and rural strata. In the second stage, **Convenience and Purposive Sampling** is deployed at specific banking touchpoints to intercept active bank account holders who possess a smartphone, ensuring the target population matches the operational requirements of the UPI ecosystem.
- **Sample Size Justification:** To move away from an unscientific "magic number" of 100, the sample size is determined using **Cochran's Sample Size Formula** for infinite or unknown populations:

$$n = \frac{(z^2 * p * q)}{e^2}$$

Where:

- n = Desired sample size.
- Z = Standard normal deviation at a 95% confidence level (1.96).
- p = Estimated proportion of an attribute present in the population (assumed at 0.50 or 50% for maximum variability).
- q = 1 - p = 0.50.
- e = Acceptable margin of error (set at 0.08 or 8%).

Calculating the parameters:

$$n = \frac{(1.96)^2 * (0.50) * (0.50)}{(0.08)^2} = \frac{3.8416 * 0.25}{0.0064} \cong 150.06$$

Note: Based on the mathematical constraint and balancing data collection viability within the regional scope, a final sample size of $N = 150$ **respondents** is established, providing a highly defensible statistical base for subsequent hypothesis testing.

7.4 Nature and Sources of Data

The study relies on a dual-data approach to ensure analytical depth:

- **Primary Data:** Gathered directly from the field using a structured, close-ended questionnaire administered online via digital forms and physically through intercept surveys at designated study settings.
- **Secondary Data:** Sourced from official statistical databases, including the National Payments Corporation of India (NPCI) monthly metrics, Reserve Bank of India (RBI) Payment System Indicators, banking annual reports, and peer-reviewed journals indexed in global databases.

7.5 Data Collection Instrument (Study Tool)

The primary research instrument is a **Structured Questionnaire** explicitly mapped to the study's objectives. It comprises three core segments:

- **Section A: Demographic Profile:** Collects nominal data regarding age, gender, education, monthly income, and occupation.
- **Section B: Awareness and Adoption Scale:** Utilizes a 5-point Likert Scale (ranging from 1 = Strongly Disagree to 5 = Strongly Agree) to measure perceived ease of use, convenience, security anxieties, and infrastructure downtime issues.
- **Section C: Operational Metrics:** Measures the actual frequency of transactions, preferred UPI service providers, and personal experiences with transaction failures.

7.6 Statistical Tools for Data Analysis

Collected data will be cleansed, coded, and analyzed using statistical software. The following analytical tools will be executed to test the hypotheses:

- **Descriptive Statistics:** Percentage analysis, mean scores, standard deviations, and frequency distribution tables to summarize demographic profiles and core adoption drivers.
- **Inferential Statistics (Hypothesis Testing): Chi-Square Test of Independence X^2 and One-Way Analysis of Variance (ANOVA)** will be executed to determine the statistical significance of relationships between independent demographic variables and dependent behavioral adoption factors.

LIMITATIONS OF THE STUDY

While this study provides valuable empirical insights into the grassroots adoption of UPI systems, certain inherent limitations must be acknowledged:

- **Geographical Restraints:** The empirical data collection is strictly confined to selected regions within the state of **Maharashtra**. Therefore, the behavioral patterns, risk perceptions, and infrastructure challenges identified here may not be entirely reflective of bank account holders in other states with different socioeconomic dynamics or digital infrastructure penetrations.
- **Sample Size Constraints:** Due to strict time allocations and resource limitations, the sample size is mathematically capped at **N = 150 respondents**. While this sample size is statistically justifiable and sufficient for executing descriptive and non-parametric inferential analyses (like Chi-Square), a larger sample across more districts would yield even more generalized conclusions.
- **Inherent Response Bias:** The primary data relies on self-reported survey responses from bank account holders. Respondents may exhibit a degree of social desirability bias (overstating their tech-savviness or digital awareness) or recall bias when reporting past transaction failures or frequency of usage.
- **Technological Exclusions:** This research focuses exclusively on the **Unified Payments Interface (UPI)** ecosystem operated via smartphones. It deliberately excludes corporate electronic funds transfer networks (like heavy-volume RTGS or corporate NEFT) and traditional non-smartphone digital banking channels, limiting its findings strictly to retail mobile payment behaviors.

CHAPTERIZATION

The final project report for this study will be organized into a structured framework consisting of five core chapters, detailed as follows:

- **Chapter 1: Introduction**

This chapter will introduce the foundational concepts of the digital payment ecosystem in India, with a specific focus on the evolution of the Unified Payments Interface (UPI). It will incorporate the **Rationale of the Study**, a comprehensive **Literature Review** tracking global and domestic technology adoption frameworks, the explicit **Objectives of the Study**, and the formulated **Hypotheses** intended for statistical validation.

- **Chapter 2: Research Methodology and Study Setting**

This chapter will provide an operational breakdown of the research implementation. It will detail the **Descriptive and Analytical Research Design**, the **Multi-Stage Stratified Random Sampling** technique, and the mathematical justification for the sample size ($N = 150$). It will also provide an in-depth profile of the urban and rural **Study Settings** in Maharashtra, alongside a breakdown of the primary data collection instrument (the structured questionnaire).

- **Chapter 3: Data Analysis and Hypothesis Testing**

The core empirical findings of the study will be presented in this chapter. It will utilize descriptive statistics (frequency distribution and percentage analysis) to present demographic profiles and core adoption drivers. Further, it will execute inferential statistical tests, including the **Chi-Square Test of Independence** and **One-Way ANOVA**, to systematically test and validate the research hypotheses.

- **Chapter 4: Summary, Findings, and Conclusion**

This chapter will synthesize the empirical data into a clear summary of key findings. It will directly map the results back to each specific research objective, concluding with an objective assessment of the current state of UPI awareness, adoption triggers, and structural resistance among bank account holders in Maharashtra.

- **Chapter 5: Recommendations and Suggestions**

Based on the empirical barriers identified in the study, this final chapter will offer actionable, strategic recommendations for commercial banks, fintech developers, and regional policymakers to bridge the digital literacy gap, minimize transaction failures, and enhance security protocols to deepen financial inclusion.

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SELF DECLARATION CERTIFICATE

I, **Abhay Shyamsundar Badwaik**, Enrollment No. **2452463586**, a student of **Master of Business Administration in Financial Management (MBAFM)** at Indira Gandhi National Open University (IGNOU), hereby solemnly declare that the Project Synopsis titled "**A Study on Customer Awareness and Adoption of UPI-Based Digital Payment Systems Among Bank Account Holders in Maharashtra**" has been prepared entirely by me through my own independent research, reading, and analysis.

I further declare that no Artificial Intelligence (AI) based tools, software, or platforms have been used at any stage in the preparation of this synopsis or project report. All content, arguments, and conclusions presented in this document are based on my own understanding, supported by referenced published sources.

I understand that if this declaration is found to be false at any point, my project may be rejected as per university policy.

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